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HA International LLC

Material Safety Data Sheet

FOR INDUSTRIAL USE ONLY

Super F® G29M1269

1. Product and company identification

Product name	Super F® G29M1269
MSDS Number	000000107062
Product Type	Resin Coated Sand
Product use	Foundry Core and Mold/Mould Applications
Manufacturer, Importer, Supplier	HA International, LLC 630 Oakmont Lane Westmont, IL 60559
Print date	17-JAN-2012
Telephone	**For Emergency Medical Assistance** Call Health & Safety Information Services, 1-866-303-6949 **For Emergency Transportation Information** CHEMTREC US Domestic (800) 424-9300 CHEMTREC International (703) 527-3887 CANUTEC CA Domestic (613) 996-6666 For additional health and safety or regulatory information, call (630)575-5722, or (630)575-5705.

2. Hazards identification

Form	Free flowing granules
Odor	Odorless/Odourless
OSHA/HCS status	This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).
Emergency overview	CAUTION ! THIS PRODUCT IS COATED WITH A PHENOLIC RESIN WHICH MAY ABRADE DURING HANDLING OR MECHANICAL CONVEYANCE GENERATING ORGANIC DUST. WHEN HANDLING OR CONVEYING IN ENCLOSED SPACES, ANY ABRADED PHENOLIC RESIN DUST FROM THIS PRODUCT CAN BE COMBUSTIBLE AND MAY PRESENT A FIRE OR EXPLOSION HAZARD WHEN DISPERSED AND IGNITED IN AIR. MAY CAUSE RESPIRATORY TRACT IRRITATION.

Potential acute health effects

Inhalation	Slightly irritating to the respiratory system.
Ingestion	Not expected to be harmful under normal conditions of use.
Skin	May cause irritation on prolonged or repeated contact.

Eyes	Exposure to airborne concentrations above statutory or recommended exposure limits may cause irritation of the eyes.

Potential chronic health effects

Chronic effects	Contains material that can cause target organ damage. Can cause fibrotic lung disease. Inhalation of silica dust may cause delayed lung injury or lung fibrosis (including silicosis and/or pneumoconiosis) and/or other diseases which may lead to permanent disability and/or death. Silicosis is a form of disabling pulmonary fibrosis which can be progressive. Prolonged exposure to respirable silica may cause diminished lung capacity with shortness of breath during physical exertion and may cause fatigue, breathlessness, wheezing, cough, and sputum production. Preexisting respiratory disorders may be aggravated by exposure. Smoking may aggravate the effects of exposure and may increase the risk of developing respiratory disease from exposure. Consult with your employer and your doctor for further information or if you believe you may be developing any breathing or lung problems. There is some evidence that breathing respirable crystalline silica or the disease silicosis is associated with an increased incidence of significant disease endpoints such as scleroderma (an immune system disorder manifested by fibrosis of the lungs, skin and other internal organs) and kidney disease. Silicosis is also reported to increase the risk of tuberculosis. Some studies show an increased incidence in chronic bronchitis and emphysema in workers exposed to respirable crystalline silica.
Carcinogenicity	Contains material which can cause cancer. Risk of cancer depends on duration and level of exposure.
Mutagenicity	No known significant effects or critical hazards.
Teratogenicity	No known significant effects or critical hazards.
Developmental effects	No known significant effects or critical hazards.
Fertility effects	No known significant effects or critical hazards.
Target organs	Contains material which causes damage to the following organs: kidneys, lungs, immune system, skin, eye, lens or cornea Review Section 2 and 11 for any additional assessments.

Note: Residual formaldehyde gas may be released from this product during processing. The amount and level will depend on local conditions of use. Formaldehyde gas is irritating to the eyes and upper respiratory tract and may aggravate existing respiratory conditions or allergies. OSHA has listed formaldehyde as a potential human carcinogen. See the OSHA formaldehyde standard 29 CFR 1910.1048 for further details. The International Agency for Research on Cancer (IARC) has classified formaldehyde as carcinogenic to humans. Product contains traces of hexamethylenetetramine which may cause allergic skin and respiratory reactions.

Over-exposure signs/symptoms

Inhalation	Adverse symptoms may include the following: respiratory tract irritation, coughing,
Ingestion	No specific data.
Skin	No specific data.
Eyes	No specific data.

Medical conditions aggravated by over-exposure Pre-existing disorders involving any target organs mentioned in this MSDS as being at risk may be aggravated by over-exposure to this product.

See section 11 for more detailed information on health effects and symptoms.

3. Composition/Information on ingredients

Ingredient name	CAS number	WT %
Quartz (SiO2)	14808-60-7	**70.0 - 100.0**
Kaolin	1332-58-7	**1.0 - 5.0**

*** Any applicable Canadian trade secret numbers will be listed in Section 15.*

4. First aid measures

Eye contact Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Check for and remove any contact lenses. Get medical attention if irritation occurs.

Skin contact Wash contaminated skin with soap and water. Get medical attention if symptoms occur.

Inhalation Get medical attention if symptoms occur. Move exposed person to fresh air. If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel.

Ingestion Get medical attention if symptoms occur. Never give anything by mouth to an unconscious person. Do not induce vomiting unless directed to do so by medical personnel.

Protection of first aid personnel No action shall be taken involving any personal risk or without suitable training. If it is suspected that fumes are still present, the rescuer should wear an appropriate mask or self-contained breathing apparatus. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

Notes to physician No specific treatment. Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.

See section 11 for more detailed information on health effects and symptoms.

5. Fire-fighting measures

Flammability of the product Not combustible per DOT 49CFR section 173 Appendix (e).

Extinguishing media

Suitable Use water.

Special exposure hazards Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.

Hazardous combustion products Decomposition products may include the following materials: carbon monoxide, carbon dioxide, smoke , soot, aromatic compounds including benzo[a]pyrene , irritating and toxic fumes and gasses .

Special protective equipment for fire-fighters Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Special Remarks on Explosion Hazards

This product is coated with a phenolic resin. The phenolic resin surface may abrade during handling or mechanical conveyance. Such abrasion may allow dispersion of organic dust. When handling or conveying in enclosed spaces, any abraded phenolic resin dust from this product can be combustible and may present a fire or explosion hazard when dispersed and ignited in air.

6. Accidental release measures

Minimize the creation and distribution of dust. Remove to a chemical disposal area. Prevent entry into natural bodies of water.

Personal precautions

No action shall be taken involving any personal risk or without suitable training. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilled material. Provide adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Put on appropriate personal protective equipment (see section 8).

Environmental precautions

Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).

Large spill

Move containers from spill area. Approach release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Vacuum material and place in a designated, labelled waste container. Dispose of via a licensed waste disposal contractor. Note: see section 1 for emergency contact information and section 13 for waste disposal.

Small spill

Move containers from spill area. Vacuum material and place in a designated, labelled waste container. Dispose of via a licensed waste disposal contractor.

7. Handling and storage

Handling

Put on appropriate personal protective equipment (see section 8). Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Workers should wash hands and face before eating, drinking and smoking. Do not get in eyes or on skin or clothing. Do not ingest. Use only with adequate ventilation. Wear appropriate respirator when ventilation is inadequate. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Empty containers retain product residue and can be hazardous. Do not reuse container.

This product is coated with a phenolic resin. The phenolic resin surface may abrade during handling or mechanical conveyance. Such abrasion may allow dispersion of organic dust. Take precautionary measures, such as capturing dust at generation points, practicing good housekeeping, controlling sources of ignition (including static electricity), and grounding equipment, to minimize dust accumulation, airborne dust, and static discharges during handling and/or conveyance as instructed in NFPA Pamphlet No. 654, "Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids," UK HSE Guidance HSG 103, approved Codes of Practice (ACOPS) established for Explosive Atmospheres under the ATEX Directive 1999/92/EC for worker protection and ATEX Directive 94/9/EC that regulates equipment and protection systems used

in potentially explosive atmospheres or other national guidance on safe handling of combustible dusts.

Storage

Store in accordance with local regulations. Store in original container protected from direct sunlight in a dry, cool and well-ventilated area, away from incompatible materials (see section 10) and food and drink. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Do not store in unlabeled containers. Use appropriate containment to avoid environmental contamination.

8. Exposure controls/personal protection

Ingredient name	Occupational exposure limits
Quartz (SiO2)	**ACGIH TLV 8-hr TWA** 0.025 mg/m3 (respirable fraction) **OSHA PEL Z3 8-hr TWA** 10 mg/m3 (divided by %SiO2+2 , respirable) **OSHA PEL Z3 8-hr TWA** 30 mg/m3 (divided by %SiO2+2 , total dust)
Kaolin	**ACGIH TLV 8-hr TWA** 2 mg/m3 (respirable fraction) **OSHA PEL 8-hr TWA** 15 mg/m3 (total dust) **OSHA PEL 8-hr TWA** 5 mg/m3 (respirable fraction)

Consult local authorities for acceptable exposure limits.

Recommended monitoring procedures

Review ASTM E 1132-99, "Standard Practice for Health Requirements Relating to Occupational Exposure to Respirable Crystalline Silica," as well as other guidelines such as NIOSH publications.

Engineering measures

Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Avoid creating dust. If airborne contaminants are generated when the material is heated, handled, or transported, appropriate administrative and engineering controls, including sufficient ventilation in volume and air flow patterns, should be provided to keep air contaminant concentration levels below acceptable criteria. Minimize the collection (build-up) of dust on walls, floors, sills, ledges, machinery, or equipment. Work clothes should be changed daily and washed to prevent dust contact. These techniques may not necessarily address all issues pertaining to your operation. We, therefore, recommend that you consult with experts of your choice to determine whether or not your programs

are adequate. Consult OSHA, state/provincial and local laws, and regulations.

Hygiene measures Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing. Ensure that eyewash stations and safety showers are close to the workstation location.

Respiratory Use a properly fitted, air-purifying or air-fed respirator complying with an approved standard if a risk assessment indicates this is necessary. Respirator selection must be based on known or anticipated exposure levels, the hazards of the product and the safe working limits of the selected respirator.

Hands Chemical-resistant, impervious gloves complying with an approved standard should be worn at all times when handling chemical products if a risk assessment indicates this is necessary.

Eyes Safety eyewear complying with an approved standard should be used when a risk assessment indicates this is necessary to avoid exposure to liquid splashes, mists or dusts.

Skin Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.

Environmental exposure controls Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation.In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

9. Physical and chemical properties

Form	Free flowing granules
Flash point	Non-flammable.
Auto-ignition temperature	Not applicable
Flammable limits	
Lower:	Not applicable
Upper:	Not applicable
Color	Tan.
Odor	Odorless/Odourless
pH	Not applicable
Boiling point	Not applicable
Relative density	Approx. 2.65
Vapor pressure	Not applicable
Odor threshold	Not applicable
Viscosity	Dynamic- Not applicable
Solubility	Negligible
Partition coefficient: n-octanol/water	Not applicable
Evaporation rate	Not applicable
Vapor density	Not applicable

No known significant effects or critical hazards.

13. Disposal considerations

Waste disposal The generation of waste should be avoided or minimized wherever possible. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

The data provided in this section is for information only and may not be specific to your package size or mode of transport. You will need to apply the appropriate regulations to properly classify your shipment for transportation.

International transport regulations

Regulatory information	UN/NA number	Proper shipping name	Classes/*PG	Reportable Quantity (RQ)
CFR		Non-regulated		
TDG		Non-regulated		
IMO/IMDG		Non-regulated		
IATA (Cargo)		Non-regulated		

*PG : Packing group

15. Regulatory information

US regulations

HCS Classification Carcinogen, Target organ effects

U.S. Federal regulations **SARA 311/312 Classification** Delayed (chronic) health hazard

SARA 313 - Supplier Notification
This product contains the following toxic chemical(s) subject to the reporting requirements of Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986, and Subpart C-Supplier Notification Requirement of 40 CFR Part 372.
None required.

SARA 302 Extremely Hazardous Substances None required.

Clean Air Act (CAA) Section 112 Hazardous Air Pollutants None required.

State regulations **Massachusetts RTK Substances** The following components are listed: Quartz (SiO_2), Kaolin,

New Jersey RTK Hazardous Substances The following components are listed: Quartz (SiO_2),

Pennsylvania RTK Hazardous Substances The following components are listed: Quartz (SiO_2), Kaolin,

California Prop. 65: WARNING: This product contains a chemical known to the State of California to cause cancer. Quartz (SiO_2) - 14808-60-7, Formaldehyde - 50-00-0,

Canada

WHMIS (Canada) Class D-2A: Material causing other toxic effects (Very toxic).

Canadian lists **Canadian NPRI:** None required.

International regulations

Chemical inventories

Europe inventory All components are listed or exempted.
Australia inventory (AICS) All components are listed or exempted.
China inventory (IECSC) All components are listed or exempted.
Japan inventory (ENCS) Not determined.
Japan inventory (ISHL) Not determined.
Korea inventory (KECI) All components are listed or exempted.
New Zealand Inventory (NZIoC) Not determined.
Philippines inventory (PICCS) All components are listed or exempted.
Canada inventory All components are listed or exempted.
United States inventory (TSCA 8b) All components are listed or exempted.

16. Other information

Hazardous Material Information System III (U.S.A.)

Health : 1
Flammability: 0
Physical hazards : 0
Chronic : *

Caution: HMIS® ratings are based on a 0-4 rating scale, with 0 representing minimal hazards or risks, and 4 representing significant hazards or risks Although HMIS® ratings are not required on MSDSs under 29 CFR 1910.1200, the preparer may choose to provide them. HMIS® ratings are to be used with a fully implemented HMIS® program. HMIS® is a registered mark of the National Paint & Coatings Association (NPCA). HMIS® materials may be purchased exclusively from J. J. Keller (800) 327-6868.
The customer is responsible for determining the PPE code for this material.

Prepared by Product Safety & Compliance Group, (630)575-5722, or (630)575-5705
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